



Olimpiada de Informatica Aplicata AcadNet 2025

Calculatoare & Sisteme de Operare

Barem Sectiunea 9–10

1. Operații cu fișiere și directoare

- (a) `mkdir -p AcadNet/{calc,is,rl}/{9-10,11-12}`
- (b) `cd AcadNet/calc/9-10`
`for i in {1..50}; do`
`dd if=/dev/zero of=subiect_$i bs=1M count=12`
`done`
- (c) `# Afișare dimensiune totală`
`du -sh AcadNet`
`# Afișare structură ierarhică`
`find AcadNet -type d -o -type f | sort`
- (d) `zip -r AcadNet.zip AcadNet`
`ls -lh AcadNet.zip`
`unzip -l AcadNet.zip`

2. Gestionarea proceselor și planificarea comenzilor

- (a) `ps -u "$USER" axo pid,%mem --sort=-%mem \`
`| head -6 | tail -5 \`
`| awk '{print $1}' \`
`| xargs renice -n 15 -p`
- (b) `echo 'echo "Hello from AcadNet 2025"' | at 00:00 01.08.2025`
- (c) `for i in {1..10}; do (sleep 30 &); done && ps -ef | grep "sleep 30" | grep -v grep`
`| awk '{print $2}' | xargs kill -9`
- (d) `exec -a "de ce rinocerul are corn?" sleep 1000 &`

3. Personalizarea terminalului

- (a) `nano /.bashrc`
decomentați linia `"#force_color_prompt=yes"` stergând `#`
cautați linia care se află sub `"if [ "$color_prompt" = yes ]; then"`
modificați astfel: `PS1='${debian_chroot:+($debian_chroot)} \[\033[01;31m\]`
`\u\[\033[01;32m\]@\[\033[01;33m \]\h\[\033[00m\]:\[\033[01;34m\]\w\[\033`
`[00m\]\$ '`
`source /.bashrc`
- (b) `cowsay -f hellokitty 'ACADNET 2025'`
- (c) `nano /.bashrc`
adăugați la final: `export PS1='\u@$(hostname -I | awk "{print \$1}"): \w\$ '`
`source /.bashrc`

- (d) nano /.bashrc
adăugați la finalul terminalului: echo -ne '\e[3 q'
source /.bashrc

4. Utilizarea Makefile

- (a) build: acadnet2025
gcc main.c firststep.c -o acadnet2025
- (b) build: acadnet2025
gcc main.c firststep.c -o acadnet2025
gcc -c secondstep.c -o secondstep.o
gcc zeus.o secondstep.o -o sample
run: acadnet2025
./acadnet2025
clean:
rm secondstep.o sample acadnet2025
- (c) build: acadnet2025
gcc main.c firststep.c -o acadnet2025
gcc -c secondstep.c -o secondstep.o
gcc zeus.o secondstep.o -o sample
run: acadnet2025
./acadnet2025
clean:
rm secondstep.o sample acadnet2025
.PHONY: pack
pack:
tar -cvzf acadnet_rules.tar.gz *.c *.h Makefile

5. Gestionarea permisiunilor

- (a) mkdir -p database/public database/secret
head -c 10M /dev/urandom > database/public/info
head -c 10M /dev/urandom > database/secret/info
- (b) tar -czvf database.tar.gz database/
tar -tzvf database.tar.gz
- (c) chmod 777 database/public
chmod 770 database/secret
touch secret/config
chown dbowner secret/config
chmod 700 secret/config
- (d) sudo groupadd dbadmin
sudo useradd -m dbowner -G dbadmin
sudo useradd -m dbuser
sudo chown :dbadmin database/secret

6. Analiza timpilor de bootare cu systemd

- (a) systemd-analyze

- (b) `systemd-analyze blame | sort -n`
- (c) `systemd-analyze blame | sort -n -u | sed 's/^*//g' | cut -d ' ' -f 2`

7. Folosirea Git

- (a) `git config --global user.name "student"`
`git config --global user.email "student@acadnet.ro"`
- (b) `mkdir acadnet2025`
`cd acadnet2025`
`git init`
`cp /etc/passwd .`
- (c) `git branch networking`
`git checkout networking`
`ifconfig > ports`
`git add .`
`git commit -m "Suspicious ports"`
- (d) `git commit --amend`
Not suspicious ports

8. Provocarea Batman vs. Joker

`sudo chmod u+s /usr/bin/id`